

Joy Dhar

DEEP LEARNING RESEARCHER | PH.D. SCHOLAR IN COMPUTER SCIENCE & ENGINEERING · IIT ROPAR

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“Building effective and reliable AI for those who need it most—not just where data is abundant, but also where care is scarce.”

Summary

Ph.D. Scholar and Deep Learning Researcher at IIT Ropar working at the intersection of Efficient AI, Multimodal AI, and Trustworthy AI for healthcare, with a focus on heterogeneous medical data, including medical imaging, multi-omics, electronic health records, and time-series signals. My work, published in top-tier venues, aims to advance scientific discovery and translate cutting-edge research into real-world clinical decision-support systems for resource-constrained medical AI applications.

Research Interests

ML Foundations: Supervised, self-supervised, unsupervised, and continual learning; distribution-shift robustness; knowledge distillation.

Multimodal AI: Multimodal representation learning and fusion; multimodal and cross-modal attention; vision–language models.

Trustworthy AI: Adversarial Attacks; Certified & Empirical Adversarial Robustness; Curriculum Learning.

Generative AI: Generative adversarial networks; diffusion models; autoencoders; variational autoencoders.

Healthcare AI: Efficient medical image analysis; heterogeneous clinical data modeling; disease classification and segmentation.

Publications/Accepted

Conference

1. **J. Dhar**, M. Pandey, B. Bozorgtabar, N. Zaidi, W. Zhang, W. H. Li, T. Mu, D. Mahapatra, M. Baktashmotlagh, T. Le, C. Chen, S. Mistry, C. Gonzalez, S. E. Kahou, L. Yao, P. Koniusz, R. B. Fisher, D. Phung, B. Han, N. Vasconcelos, and P. Lio, “Does a Hybrid Space-Aware Randomized Defense Improve Empirical and Certified Adversarial Robustness?,” *ICML*, 2026.
2. **J. Dhar**, S. Xia, M. Pandey, M. Haghighat, A. Alavi, F. Sohel, W. Zhang, and N. Zaidi, “Certified VS. Empirical adversarial robustness via hybrid convolutions with attention stochasticity,” *ICLR*, 2026.
3. **J. Dhar**, N. Zaidi, M. Haghighat, “Effective and Robust Multimodal Medical Image Analysis,” *ACM SIGKDD Conference (Research Track), Oral Presentation*, 2026.
4. **J. Dhar**, M. Pandey, N. Zaidi, C. Chen, M. Haghighat, F. Sohel, and P. Goyal, “Advancing Multimodal Fusion on Heterogeneous Medical Data with Hybrid Geometry Attention,” *ACM SIGKDD Conference (Research Track)*, 2026 (Accepted).
5. D. Mahapatra, A. Das, M. Pandey, **J. Dhar**, S. Roy, Z. Ge, B. Bozorgtabar, and I. Razzak, “Proximity-Constrained Counterfactual Decoding for Hallucination-Robust Medical VQA,” *ECCV*, 2026 (Accepted).
6. D. Mahapatra, A. Das, S. Roy, M. Pandey, **J. Dhar**, Z. Ge, B. Bozorgtabar, and I. Razzak, “Prompt and Probe: Data-Efficient Adaptation of Black-Box Foundation Models via Unified Active Visual Prompting,” *MICCAI*, 2026 (Accepted).
7. **J. Dhar**, M. Pandey, D. Chakladar, M. Haghighat, A. Alavi, S. Mistry, N. Zaidi, “HyPCA-Net: Advancing Multimodal Fusion in Medical Image Analysis,” *WACV*, 2026.
8. **J. Dhar**, N. Zaidi, M. Haghighat, S. Roy, P. Goyal, A. Alavi, and V. Kumar, “Multimodal fusion learning with dual attention for medical imaging,” *WACV*, 2025.
9. **J. Dhar**, K. Rana, and P. Goyal, “Uncertainty-aware robust information fusion attention network for brain tumors classification in MRI images,” *ICPR*, 2024.
10. M. S. Kanroo, H. S. Kawoosa, **J. Dhar**, and P. Goyal, “PiExtract: an end-to-end data extraction pipeline for pie charts,” *ICPR*, 2024.

Journal

1. **J. Dhar**, P. Goyal, M. Haghighat, N. Zaidi, F. Sohel, Q. B. Vo, and K. C. Santosh, “Towards building robust models for unimodal and multimodal medical imaging data,” *Information Fusion*, 2026.
2. **J. Dhar**, K. Rana, P. Goyal, A. Alavi, R. Rana, Q. B. Vo, S. Mishra, and S. Mistry, “Dual-phase neural networks for feature extraction and ensemble learning for recognizing human health activities,” *Applied Soft Computing*, 2025.
3. **J. Dhar**, and S. Roy, “Identification and diagnosis of cervical cancer using a hybrid feature selection approach with the Bayesian optimization– based optimized CatBoost classification algorithm,” *Journal of Ambient Intelligence and Humanized Computing*, 2024.
4. **J. Dhar**, “An adaptive intelligent diagnostic system to predict early stage of Parkinson’s disease using two-stage dimension reduction with genetically optimized LightGBM algorithm,” *Neural Computing & Applications*, 2022.
5. **J. Dhar** and N. A. Ayele, “Multi-tier ensemble learning model with neighborhood component analysis to predict health diseases,” *IEEE Access*, 2021.
6. **J. Dhar**, “Multistage ensemble learning model with weighted voting and genetic algorithm optimization strategy for detecting chronic obstructive pulmonary disease,” *IEEE Access*, 2021.
7. **J. Dhar**, A. Jodder, “An Effective Recommendation System to Forecast the Best Educational Program Using Machine Learning Classification Algorithms,” *Ingénierie des Systèmes d’Information*, 2020.

Manuscripts Under Review/Preparation

Conference

1. **J. Dhar**, M. Pandey, P. Goyal, D. Mahapatra, R. Narava, N. Zaidi, B. Bozorgtabar, and P. Lio, "CoVANT: Advancing Multimodal Fusion on Heterogeneous Medical Data," submitted to **NeurIPS 2026**, manuscript under review.
2. **J. Dhar**, M. Pandey, P. Goyal, D. Mahapatra, R. Narava, N. Zaidi, B. Bozorgtabar, Z. Ge, P. Koniusz, P. Lio, M. Fiaz, and N. Vasconcelos, "GSAND: Hybrid-Space Attention Advances Medical Image Segmentation," submitted to **NeurIPS 2026**, manuscript under review.
3. **J. Dhar**, M. Pandey, P. Goyal, D. Mahapatra, B. Bozorgtabar, S. Biswas, N. Tiwai, H. Müller, M. Fiaz, B. Han, P. Koniusz, and P. Lio, "Certified vs. Empirical Adversarial Robustness in Multimodal Fusion via Hybrid Space Convolutional Stochasticity," submitted to **NeurIPS 2026**, manuscript under review.
4. D. Mahapatra, A. Das, **J. Dhar**, M. Pandey, Z. Ge, B. Bozorgtabar, and I. Razzak, "Transductive Conformal Decoding: Controlling Hallucinations in Multimodal Generation," submitted to **NeurIPS 2026**, manuscript under review.
5. **J. Dhar**, D. Chakladar, M. Pandey, D. Mahapatra, N. Zaidi, B. Bozorgtabar, I. Razzak, H. Mokayed, P. Goyal, "GCARE: Advancing Multimodal Fusion on Heterogeneous Medical Data," manuscript in preparation for submission to **KDD 2027**.
6. **J. Dhar**, R. Goyal, P. Goyal, "Can Robust Gradient Attacks Reveal Vulnerabilities in Deep Networks?," manuscript in preparation for submission to **KDD 2027**.
7. **J. Dhar**, M. Pandey, P. Goyal, N. Zaidi, W. H. Li, B. Bozorgtabar, D. Mahapatra, T. Mu, T. V. Nguyen, C. Sun, E. Yang, N. Aghaeepour, M. Baktashmotlagh, D. Phung, R.B. Fisher, C. Davatzikos, "DEMAND: Redefining Decoder Design to Advance Medical Image Segmentation," manuscript in preparation for submission to **AAAI 2027**.
8. **J. Dhar**, M. Pandey, R. Narava, P. Goyal, D. Mahapatra, P. Koniusz, N. Zaidi, I. Razzak, B. Bozorgtabar, C. Sun, J. Yang, M. Reyes, Z. Ge, L. Yao, M. Baktashmotlagh, A. Farshad, I. Banerjee, W. Zhang, M. Fiaz, C. Gonzalez, E. Yang, H. Müller, K. Åström, N. Vasconcelos, P. Lio, "Can Multimodal Fusion of Heterogeneous Medical Data Accelerate Convergence?," manuscript in preparation for submission to **ICLR 2027**.
9. **J. Dhar**, M. Pandey, P. Goyal, B. Bozorgtabar, D. Mahapatra, P. Koniusz, B. Han, N. Vasconcelos, P. Lio, "Can Holonomy-Guided Attacks Expose Geometric Vulnerabilities in Deep Networks?," manuscript in preparation for submission to **ICLR 2027**.
10. **J. Dhar**, M. Pandey, P. Goyal, B. Bozorgtabar, D. Mahapatra, P. Koniusz, B. Han, N. Vasconcelos, P. Lio, "Can Diffusion-Guided Attacks Uncover Vulnerabilities in Deep Networks?," manuscript in preparation for submission to **ICLR 2027**.
11. **J. Dhar**, M. Pandey, P. Goyal, B. Bozorgtabar, D. Mahapatra, P. Koniusz, B. Han, N. Vasconcelos, P. Lio, "Can Deterministic-Stochastic Learning Bridge Certified and Empirical Adversarial Robustness?," manuscript in preparation for submission to **ICLR 2027**.

Journal

1. **J. Dhar**, M. Pandey, P. Goyal, S. Biswas, P. Koniusz, N. Vasconcelos, and P. Lio, "Can Hybrid-Space Representation Learning Advance Multimodal Fusion?," manuscript in preparation for submission to *Transactions on Machine Learning Research (TMLR)*.
2. **J. Dhar**, P. Goyal, and F. Sohel, "MANAT: Multimodal attention noise injection with adversarial training for adversarial robustness in medical image analysis," submitted to *Applied Soft Computing*, manuscript under review.

Education

Indian Institute of Technology Ropar (IIT Ropar)

PH.D. IN COMPUTER SCIENCE AND ENGINEERING

- Advisor: Dr. Puneet Goyal; Institute Scholarship Recipient

Punjab, India

Jul. 2022--Jun. 2027

Maulana Abul Kalam Azad University of Technology

M.TECH. IN COMPUTER SCIENCE AND ENGINEERING

- CGPA 8.92/10 (A+); ranked 3rd in the department

West Bengal, India

Aug. 2015--Nov. 2017

West Bengal University of Technology

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING

- CGPA 8.59/10 (A+); ranked 10th in the department

West Bengal, India

Aug. 2009--Nov. 2012

Awards & Honors

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|------|---|------------------|
| 2025 | Recipient , Anusandhan National Research Foundation (ANRF) Travel Fund to present at WACV 2025 | Tucson, USA |
| 2024 | Winner , IEEE Video and Image Processing (VIP) Cup at ICIP 2024 | Abu Dhabi, UAE |
| 2011 | Finalist , IBM India The Great Mind Challenge (TGMC) Project | Bangalore, India |

Scholarships/Fellowships

INTERNATIONAL

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|------|--|-----------|
| 2022 | Deakin University Postgraduate Research Scholarship , Deakin University | Australia |
| 2022 | Industrial Scholarship for Ph.D., Big Data and High-Performance Computing , University of Genoa | Italy |
| 2022 | Ph.D. Institute Scholarship , University of Salerno | Italy |
| 2022 | JCU Postgraduate Research Stipend Scholarship , James Cook University | Australia |
| 2021 | Ph.D. Institute Scholarship , National Taipei University of Technology | Taiwan |
| 2021 | Ph.D. Institute Scholarship , National Dong Hwa University | Taiwan |
| 2021 | Elite Doctoral Scholarship , National Sun Yat-sen University | Taiwan |

DOMESTIC

- 2022 **Ph.D. Institute Scholarship**, Indian Institute of Technology Dhanbad
- 2022 **Ph.D. Institute Scholarship**, Indian Institute of Technology Ropar
- 2019 **NET Fellowship**, NTA-UGC
- 2018 **GATE Fellowship**, M.H.R.D.

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Current Projects

Does a Hybrid Space-Aware Weight Refiner Improve Multimodal Fusion Robustness?

Punjab, India

DEVISING HYBRID SPACE AWARE ATTENTION FOR MULTIMODAL FUSION

June, 2026

- Devised a hybrid space aware weight refinement module that incorporates both implicit and explicit refinement process to improve multimodal fusion robustness.

Robust-Zero: A Defense Framework for Zero-Shot Classification

Punjab, India

DEVELOPING A ZERO-SHOT DEFENSE FRAMEWORK USING STOCHASTIC COMPONENTS AND FEATURE DISENTANGLEMENT

June 2025

- Jointly exploit random filtering, attention-noise injection, and disentangled feature representations to mitigate vulnerabilities.

Academic Services

- 2021–2026 **PC member/Reviewer**, NeurIPS; CVPR; AAAI; KDD; ACM MM; WACV; ICPR; ASOC; NCAA; Scientific Reports
- 2022–2027 **Teaching Assistant**, Indian Institute of Technology Ropar
- 2021–2022 **Research Assistant**, National Sun Yat-sen University
- 2016–2021 **Information Technology Teacher**, Hatgobindapur M. C. High School
- 2014–2016 **Lecturer in Computer Science & Technology**, JLD College of Engineering & Management

India & Taiwan
Punjab, India
Kaohsiung, Taiwan
Burdwan, India
Kolkata, India

Technical Skills

Programming: Python, C/C++, Java, C#, PHP, MATLAB;

Frameworks: TensorFlow/Keras, PyTorch, scikit-learn;

Data Analysis: NumPy, Pandas, OpenCV, SciPy;

Tools & Platforms: LaTeX, Jupyter, Kaggle, Google Colab, VS Code.

Selected Coursework

Deep Learning
Internet of Things

Computer Vision
Pattern Recognition

Advanced Computer Vision
Artificial Intelligence

Digital Image Processing
Data Structures & Algorithms

References

Dr. Puneet Goyal

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Full Professor (Co-Author)
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